

Workshop - Understanding Bank Financial Statements

AFIN8003 Workshop

Dr. Mingze Gao

2026-08-06

What bank accounting will get you

Understanding bank accounting is fundamental for anyone interested in finance, banking, or risk management. Banks play a unique role in the financial system, and their accounting practices differ significantly from those of non-financial firms. Through this workshop, you will:

- Gain practical insights into how banks record and report financial transactions.
- Learn how various banking activities impact the balance sheet and financial health of a bank.
- Build a strong foundation for more advanced topics in banking, regulation, and risk management.

This workshop is based on the [BRMS \(Bank Risk Management Simulation\)](#) Python package, which I developed for teaching and research in banking.

Real-world example

CBA's balance sheet and income statement

Let's see the statements of an Australian bank, [Commonwealth Bank of Australia \(CBA\)](#), as an example. Its financial results are publicly available at [its website](#). However, to make our lives a bit easier, pre-compiled data are available on iLearn.

- Visit AFIN8003 course page on [iLearn](#).
- Under Week 1, find and download the Excel spreadsheet `SPGlobal_CBA_BalanceSheet.xlsx` and `SPGlobal_CBA_IncomeStatement.xlsx`.

i Note

Now, use the provided data and your own research, try to answer the following questions. Unless specified otherwise, use the bank's latest data.

Some (motivating) questions

1. Looking at a bank's balance sheet, what do you think a bank actually *does* as a business?
2. Why are customer deposits recorded as *liabilities* rather than as income?
3. When a bank makes a loan and holds it, what do you think happens in its accounting records and financial statements?

Simple banking book accounting

A first look at bank's balance sheet

To begin with, a simple bank has just been established with shareholders' contribution of \$10,000.

```
import datetime
from brms.accounting.account import AccountBalances
from brms.accounting.report import Report
from brms.accounting.statement_viewer import HTMLStatementViewer
from brms.models.bank import Bank

bank = Bank()
balances = AccountBalances(
    {
        bank.chart_of_accounts.cash_account: 10_000,
        bank.chart_of_accounts.equity_account: 10_000,
    },
)
bank.initialize(balances)
viewer = HTMLStatementViewer(jupyter=True, hide_zero_balance_accounts=True)

today = datetime.date(2026, 8, 6)
report = Report(ledger=bank.ledger, viewer=viewer, date=today)
```

```
report.print_balance_sheet()
```

Balance Sheet

Assets	
Cash and Cash Equivalents	\$10,000.00
Total assets	\$10,000.00
Liabilities	
Total liabilities	\$0.00
Net assets	\$10,000.00
Shareholders' equity	
Shareholders' Equity	\$10,000.00
Total shareholders' equity	\$10,000.00

Date: 2026-08-06

Customer deposit

Now some customers have made some deposits, for example:

- Customer A deposits \$60,000.
- Customer B deposits \$40,000.
- Customer C deposits \$20,000.

Notably, customer deposits are the bank's *liability*. We see an increase in both the bank's total assets and total liabilities, while total shareholders' equity remains unchanged.

```
from brms.instruments.factory import InstrumentFactory
from brms.models.transaction import TransactionFactory, TransactionType

customer_A_deposit = InstrumentFactory.create_deposit(value=60_000)
customer_B_deposit = InstrumentFactory.create_deposit(value=40_000)
customer_C_deposit = InstrumentFactory.create_deposit(value=20_000)
tx_deposit1 = TransactionFactory.create_transaction(
    bank=bank,
    transaction_type=TransactionType.DEPOSIT_RECEIVED,
    instrument=customer_A_deposit,
    transaction_date=today,
```

```

        description="Customer A's Deposits",
    )
    tx_deposit2 = TransactionFactory.create_transaction(
        bank=bank,
        transaction_type=TransactionType.DEPOSIT_RECEIVED,
        instrument=customer_B_deposit,
        transaction_date=today,
        description="Customer B's Deposits",
    )
    tx_deposit3 = TransactionFactory.create_transaction(
        bank=bank,
        transaction_type=TransactionType.DEPOSIT_RECEIVED,
        instrument=customer_C_deposit,
        transaction_date=today,
        description="Customer C's Deposits",
    )
    bank.process_transaction(tx_deposit1)
    bank.process_transaction(tx_deposit2)
    bank.process_transaction(tx_deposit3)

    report = Report(ledger=bank.ledger, viewer=viewer, date=today)
    report.print_balance_sheet()

```

Balance Sheet

Assets	
Cash and Cash Equivalents	\$130,000.00
Total assets	\$130,000.00
 Liabilities	
Deposits and Other Public Borrowings	\$120,000.00
Deposits	\$120,000.00
Total liabilities	\$120,000.00
 Net assets	 \$10,000.00
 Shareholders' equity	
Shareholders' Equity	\$10,000.00
Total shareholders' equity	\$10,000.00

Date: 2026-08-06

Deposit withdraw

Depositors can also withdraw from the bank.

If Customer C has withdrawn all of their deposits (\$20,000), we see a decrease of cash and deposits.

```
from brms.models.transaction import DepositWithdrawTransaction

tx_deposit_withdraw = TransactionFactory.create_transaction(
    bank=bank,
    transaction_type=TransactionType.DEPOSIT_WITHDRAWAL,
    instrument=customer_C_deposit,
    transaction_date=today,
    description="Customer C withdraw",
)
bank.process_transaction(tx_deposit_withdraw)

report = Report(ledger=bank.ledger, viewer=viewer, date=today)
report.print_balance_sheet()
```

Balance Sheet

Assets	
Cash and Cash Equivalents	\$110,000.00
Total assets	\$110,000.00
Liabilities	
Deposits and Other Public Borrowings	\$100,000.00
Deposits	\$100,000.00
Total liabilities	\$100,000.00
Net assets	\$10,000.00
Shareholders' equity	
Shareholders' Equity	\$10,000.00
Total shareholders' equity	\$10,000.00

Date: 2026-08-06

Loan disbursement

Of course, the typical business of a bank is to make loans. Let's assume the bank has made an (interest-only) loan of \$80,000 to some borrowers.

```
from brms.instruments.base import Instrument
from brms.instruments.mock import MockInstrument
from brms.models.transaction import LoanDisbursementTransaction

(loan := MockInstrument("A loan")).value = 80_000
# fmt: off
tx_loan = TransactionFactory.create_transaction(
    bank=bank,
    transaction_type=TransactionType.LOAN_DISBURSEMENT,
    instrument=loan,
    transaction_date=today,
    description="Loan made to a borrower",
)
# fmt: on
bank.process_transaction(tx_loan)

report = Report(ledger=bank.ledger, viewer=viewer, date=today)
report.print_balance_sheet()
```

Balance Sheet

Assets	
Cash and Cash Equivalents	\$30,000.00
Loans and Advances	\$80,000.00
Total assets	\$110,000.00
Liabilities	
Deposits and Other Public Borrowings	\$100,000.00
Deposits	\$100,000.00
Total liabilities	\$100,000.00
Net assets	\$10,000.00
Shareholders' equity	
Shareholders' Equity	\$10,000.00
Total shareholders' equity	\$10,000.00

Interest paid on deposits

After some time (e.g., a month), the bank may need to pay interest on customers' deposits, e.g., \$100.

Tip

Income Statement is included given that we'd like to know the profit and loss (P&L) of the bank over the period.

```
from brms.instruments.cash import Cash
from brms.models.transaction import InterestPaidOnDepositTransaction

today = today + datetime.timedelta(days=31)

tx_interest_paid = TransactionFactory.create_transaction(
    bank=bank,
    transaction_type=TransactionType.INTEREST_PAID_ON_DEPOSIT,
    instrument=Cash(value=100),
    transaction_date=today,
    description="Interest paid on deposits",
)
bank.process_transaction(tx_interest_paid)

report = Report(ledger=bank.ledger, viewer=viewer, date=today)
report.print_balance_sheet()
report.print_income_statement()
```

Balance Sheet

Assets

Cash and Cash Equivalents	\$29,900.00
Loans and Advances	\$80,000.00
Total assets	\$109,900.00

Liabilities

Deposits and Other Public Borrowings	\$100,000.00
Deposits	\$100,000.00
Total liabilities	\$100,000.00

Net assets	\$9,900.00
Shareholders' equity	
Shareholders' Equity	\$10,000.00
Retained Earnings	(\$100.00)
Total shareholders' equity	\$9,900.00

Date: 2026-09-06

Income Statement

Income	\$0.00
Expense	\$100.00
Interest Expense	\$100.00
Profit	(\$100.00)

Date: 2026-09-06

Loan interest income

Note

If the bank has only paid interest on deposits over this period, it surely has a net loss or negative profit as shown before.

However, it is more reasonable that the bank also has earned some profits during the same time, especially because it has made some loans.

Suppose that over the 1-month period, the bank has earned some interest income from the loans made earlier, e.g., for a total of \$800.

Tip

Overall, the bank has a net profit of \$700. We can observe that the bank's **Retained Earnings** has increased by \$700, too.

```
tx_interest_earned = TransactionFactory.create_transaction(
    bank=bank,
    transaction_type=TransactionType.LOAN_INTEREST_PAYMENT,
```

```

        instrument=Cash(value=800),
        transaction_date=today,
        description="Interest earned from loans",
    )
    bank.process_transaction(tx_interest_earned)

    report = Report(ledger=bank.ledger, viewer=viewer, date=today)
    report.print_balance_sheet()
    report.print_income_statement()

```

Balance Sheet

Assets	
Cash and Cash Equivalents	\$30,700.00
Loans and Advances	\$80,000.00
Total assets	\$110,700.00
Liabilities	
Deposits and Other Public Borrowings	\$100,000.00
Deposits	\$100,000.00
Total liabilities	\$100,000.00
Net assets	\$10,700.00
Shareholders' equity	
Shareholders' Equity	\$10,000.00
Retained Earnings	\$700.00
Total shareholders' equity	\$10,700.00

Date: 2026-09-06

Income Statement

Income	\$800.00
Interest Income	\$800.00
Expense	\$100.00
Interest Expense	\$100.00
Profit	\$700.00

Date: 2026-09-06

Intermediate trading book accounting

Trading Book

The transactions and non-transactions previous discussed all occur to the bank's **banking book**.

However, a bank also has a **trading book** that holds securities for short-term trading P&L purposes.

Note

These instruments are named **Fair Value Through Profit and Loss (FVTPL)**.

Accounting for trading book securities is relatively easy. We consider:

1. The purchase
2. The marking-to-market
3. The sale

FVTPL security purchase

Assume that the bank just purchased a FVTPL security for its trading book at its fair value of \$10,000.

The fair value of the security is recorded on the balance sheet under the *Assets at FVTPL* account.

```
report1 = Report(ledger=bank.ledger, viewer=viewer, date=today)
report1.print_balance_sheet()
```

Balance Sheet

Assets

Cash and Cash Equivalents	\$30,700.00
Loans and Advances	\$80,000.00
Total assets	\$110,700.00

Liabilities

Deposits and Other Public Borrowings	\$100,000.00
Deposits	\$100,000.00
Total liabilities	\$100,000.00

Net assets	\$10,700.00
Shareholders' equity	
Shareholders' Equity	\$10,000.00
Retained Earnings	\$700.00
Total shareholders' equity	\$10,700.00

Date: 2026-09-06

```

from brms.models.base import BookType

fvtpl_security = MockInstrument("A FVTPL Security", BookType.TRADING_BOOK)
fvtpl_security.value = fvtpl_security_value
tx_fvtpl_purchase = TransactionFactory.create_transaction(
    bank=bank,
    transaction_type=TransactionType.SECURITY_PURCHASE_TRADING,
    instrument=fvtpl_security,
    transaction_date=today,
    description="Purchase a FVTPL security",
)
bank.process_transaction(tx_fvtpl_purchase)

report2 = Report(ledger=bank.ledger, viewer=viewer, date=today)
report2.print_balance_sheet()

```

Balance Sheet

Assets	
Cash and Cash Equivalents	\$20,700.00
Loans and Advances	\$80,000.00
Assets at FVTPL	\$10,000.00
Total assets	\$110,700.00
Liabilities	
Deposits and Other Public Borrowings	\$100,000.00
Deposits	\$100,000.00
Total liabilities	\$100,000.00
Net assets	\$10,700.00
Shareholders' equity	

Shareholders' Equity	\$10,000.00
Retained Earnings	\$700.00
Total shareholders' equity	\$10,700.00

Date: 2026-09-06

FVTPL security marking to market

- The *fair value* of FVTPL security is **marked to market** periodically, e.g., daily.
- This ensures consistency in that they are held for trading P&L in the first place.

There are two possible cases:

1. Unrealized trading loss
2. Unrealized trading gain

Note

These trading gains and losses are *unrealized* because the securities have not yet been sold.

Unrealized trading loss - Income Statement

If the market value of the security went down by 5% (\$500.0) the next day, we need to recognize this *unrealized trading loss* in the Income Statement.

```
report2.print_income_statement()
```

Income Statement

Income	\$800.00
Interest Income	\$800.00
Expense	\$100.00
Interest Expense	\$100.00
Profit	\$700.00

Date: 2026-09-06

```

from brms.instruments.mock import MockValuationVisitor

today = today + datetime.timedelta(days=1)
# Assume that the market value of the FVTPL security went down by 5%
new_fvtpl_instrument_value = fvtpl_security.value * 0.95
valuation_visitor = MockValuationVisitor(new_value=new_fvtpl_instrument_value)
tx_fvtpl_marked_down = TransactionFactory.create_transaction(
    bank=bank,
    transaction_type=TransactionType.SECURITY_FVTPL_MARK_TO_MARKET,
    instrument=fvtpl_security,
    transaction_date=today,
    valuation_visitor=valuation_visitor,
    description="Mark to market a FVTPL security (down)",
)
bank.process_transaction(tx_fvtpl_marked_down)

report3 = Report(ledger=bank.ledger, viewer=viewer, date=today)
report3.print_income_statement()

```

Income Statement

Income	\$300.00
Interest Income	\$800.00
Trading Income (FVTPL)	(\$500.00)
Unrealized Trading Loss	\$500.00
Expense	\$100.00
Interest Expense	\$100.00
Profit	\$200.00

Date: 2026-09-07

Unrealized trading loss - Balance Sheet

If we were to generate the balance sheet now, such net trading loss would reduce the bank's retained earnings.

```
report2.print_balance_sheet()
```

Balance Sheet

Assets	
Cash and Cash Equivalents	\$20,700.00
Loans and Advances	\$80,000.00
Assets at FVTPL	\$10,000.00
Total assets	\$110,700.00
Liabilities	
Deposits and Other Public Borrowings	\$100,000.00
Deposits	\$100,000.00
Total liabilities	\$100,000.00
Net assets	\$10,700.00
Shareholders' equity	
Shareholders' Equity	\$10,000.00
Retained Earnings	\$700.00
Total shareholders' equity	\$10,700.00

Date: 2026-09-06

```
report3.print_balance_sheet()
```

Balance Sheet

Assets	
Cash and Cash Equivalents	\$20,700.00
Loans and Advances	\$80,000.00
Assets at FVTPL	\$9,500.00
Total assets	\$110,200.00
Liabilities	
Deposits and Other Public Borrowings	\$100,000.00
Deposits	\$100,000.00
Total liabilities	\$100,000.00
Net assets	\$10,200.00
Shareholders' equity	
Shareholders' Equity	\$10,000.00

Retained Earnings	\$200.00
Total shareholders' equity	\$10,200.00

Date: 2026-09-07

! Important

Usually, financial reports are generated at a period's end, such as quarter's end or year's end. What's shown here should be viewed as *what if we close the bank's accounting ledger at this time and generate the reports?*

Unrealized trading gain - Income Statement

The day after, if the market value of the security went up by 10% (\$950.0), we need to recognize this *unrealized trading gain*.

```
report3.print_income_statement()
```

Income Statement

Income	\$300.00
Interest Income	\$800.00
Trading Income (FVTPL)	(\$500.00)
Unrealized Trading Loss	\$500.00
Expense	\$100.00
Interest Expense	\$100.00
Profit	\$200.00

Date: 2026-09-07

```
today = today + datetime.timedelta(days=1)
# Assume that the market value of the FVTPL security went up by 10%
new_fvtpl_instrument_value = fvtpl_security.value * (1 + day2_pct_change / 100)
valuation_visitor = MockValuationVisitor(new_value=new_fvtpl_instrument_value)
tx_fvtpl_marked_up = TransactionFactory.create_transaction(
    bank=bank,
    transaction_type=TransactionType.SECURITY_FVTPL_MARK_TO_MARKET,
    instrument=fvtpl_security,
```

```

        transaction_date=today,
        valuation_visitor=valuation_visitor,
        description="Mark to market a FVTPL security (up)",
    )
    bank.process_transaction(tx_fvtpl_marked_up)

    report4 = Report(ledger=bank.ledger, viewer=viewer, date=today)
    report4.print_income_statement()

```

Income Statement

Income	\$1,250.00
Interest Income	\$800.00
Trading Income (FVTPL)	\$450.00
Unrealized Trading Gain	\$950.00
Unrealized Trading Loss	\$500.00
 Expense	 \$100.00
Interest Expense	\$100.00
 Profit	 \$1,150.00

Date: 2026-09-08

Unrealized trading gain - Balance Sheet

Again, if we were to close the bank's accounting ledger now and check its balance sheet, we will see changes of the bank's retained earnings (increase by \$950.0).

```
report3.print_balance_sheet()
```

Balance Sheet

Assets	
Cash and Cash Equivalents	\$20,700.00
Loans and Advances	\$80,000.00
Assets at FVTPL	\$9,500.00
Total assets	\$110,200.00
 Liabilities	

Deposits and Other Public Borrowings	\$100,000.00
Deposits	\$100,000.00
Total liabilities	\$100,000.00
Net assets	\$10,200.00
Shareholders' equity	
Shareholders' Equity	\$10,000.00
Retained Earnings	\$200.00
Total shareholders' equity	\$10,200.00

Date: 2026-09-07

report4.print_balance_sheet()

Balance Sheet

Assets	
Cash and Cash Equivalents	\$20,700.00
Loans and Advances	\$80,000.00
Assets at FVTPL	\$10,450.00
Total assets	\$111,150.00
Liabilities	
Deposits and Other Public Borrowings	\$100,000.00
Deposits	\$100,000.00
Total liabilities	\$100,000.00
Net assets	\$11,150.00
Shareholders' equity	
Shareholders' Equity	\$10,000.00
Retained Earnings	\$1,150.00
Total shareholders' equity	\$11,150.00

Date: 2026-09-08

FVTPL security sale

The next day, if the bank sold the security at the prevailing market price (assumed 2% up), we recognize *realized trading gain*.

Tracking the P&L of this particular FVTPL security,

1. On day 1, we had a loss of \$500.0 ($\$10000 * 95\% = \9500.0).
2. On day 2, we had a gain of \$950.0 ($\$9500.0 * 110\% = \10450.0).
3. On day 3, we had a gain of \$209.0 ($\$10450.0 * 102\% = \10659.0).

The total P&L from this security is therefore a net gain of \$659.0, which shows up in the Income Statement.

i Note

Because of the sale, the *unrealized trading gain/loss* associated with this security must be **reclassified** to *realized gain/loss*.

FVTPL security sale - Income Statement

```
report4.print_income_statement()
```

Income Statement

Income	\$1,250.00
Interest Income	\$800.00
Trading Income (FVTPL)	\$450.00
Unrealized Trading Gain	\$950.00
Unrealized Trading Loss	\$500.00
Expense	\$100.00
Interest Expense	\$100.00
Profit	\$1,150.00

Date: 2026-09-08

```
today = today + datetime.timedelta(days=1)
# Assume that the market value of the FVTPL security went up by 2%
new_fvtpl_instrument_value = fvtpl_security.value * (1 + day3_pct_change / 100)
valuation_visitor = MockValuationVisitor(new_value=new_fvtpl_instrument_value)
tx_fvtpl_marked_up = TransactionFactory.create_transaction(
    bank=bank,
    transaction_type=TransactionType.SECURITY_FVTPL_MARK_TO_MARKET,
```

```

        instrument=fvtpl_security,
        transaction_date=today,
        valuation_visitor=valuation_visitor,
        description="Mark to market a FVTPL security (up)",
    )
    bank.process_transaction(tx_fvtpl_marked_up)
    # Sale transaction should be created after the previous transaction has been processed
    tx_fvtpl_sale = TransactionFactory.create_transaction(
        bank=bank,
        transaction_type=TransactionType.SECURITY_SALE_TRADING,
        instrument=fvtpl_security,
        transaction_date=today,
        valuation_visitor=valuation_visitor,
        description="Sale of FVTPL security (net gain)",
    )
    bank.process_transaction(tx_fvtpl_sale)
    # fmt: on

    report5 = Report(ledger=bank.ledger, viewer=viewer, date=today)
    report5.print_income_statement()
    # report.print_balance_sheet()

```

Income Statement

Income	\$1,459.00
Interest Income	\$800.00
Trading Income (FVTPL)	\$659.00
Realized Trading Gain	\$1,159.00
Realized Trading Loss	\$500.00
Expense	\$100.00
Interest Expense	\$100.00
Profit	\$1,359.00

Date: 2026-09-09

- Trading gain: $\$209.0 + \$950.0 = \$1159.0$
- Trading loss: $\$500.0$
- Net gain: $\$659.0$
- Gain/loss *reclassified* from unrealized to realized

FVTPL security sale - Balance Sheet

Now, if we were to close the bank's accounting ledger and produce its balance sheet, we will have:

```
report4.print_balance_sheet()
```

Balance Sheet

Assets

Cash and Cash Equivalents	\$20,700.00
Loans and Advances	\$80,000.00
Assets at FVTPL	\$10,450.00
Total assets	\$111,150.00

Liabilities

Deposits and Other Public Borrowings	\$100,000.00
Deposits	\$100,000.00
Total liabilities	\$100,000.00

Net assets	\$11,150.00
------------	-------------

Shareholders' equity

Shareholders' Equity	\$10,000.00
Retained Earnings	\$1,150.00
Total shareholders' equity	\$11,150.00

Date: 2026-09-08

```
report5.print_balance_sheet()
```

Balance Sheet

Assets

Cash and Cash Equivalents	\$31,359.00
Loans and Advances	\$80,000.00
Total assets	\$111,359.00

Liabilities

Deposits and Other Public Borrowings	\$100,000.00
Deposits	\$100,000.00

Total liabilities	\$100,000.00
Net assets	\$11,359.00
Shareholders' equity	
Shareholders' Equity	\$10,000.00
Retained Earnings	\$1,359.00
Total shareholders' equity	\$11,359.00

Date: 2026-09-09

FVTPL security sale - Balance Sheet

Compared to the balance sheet before the bank purchased this FVTPL security, the retained earnings increased by exactly \$659.0.

```
report1.print_balance_sheet()
```

Balance Sheet

Assets	
Cash and Cash Equivalents	\$30,700.00
Loans and Advances	\$80,000.00
Total assets	\$110,700.00
Liabilities	
Deposits and Other Public Borrowings	\$100,000.00
Deposits	\$100,000.00
Total liabilities	\$100,000.00
Net assets	\$10,700.00
Shareholders' equity	
Shareholders' Equity	\$10,000.00
Retained Earnings	\$700.00
Total shareholders' equity	\$10,700.00

Date: 2026-09-06

```
report5.print_balance_sheet()
```

Balance Sheet

Assets	
Cash and Cash Equivalents	\$31,359.00
Loans and Advances	\$80,000.00
Total assets	\$111,359.00
Liabilities	
Deposits and Other Public Borrowings	\$100,000.00
Deposits	\$100,000.00
Total liabilities	\$100,000.00
Net assets	\$11,359.00
Shareholders' equity	
Shareholders' Equity	\$10,000.00
Retained Earnings	\$1,359.00
Total shareholders' equity	\$11,359.00

Date: 2026-09-09

Intermediate banking book accounting

Banking book

Apart from loans, a bank's banking book contains other assets that are held mostly for interest income. However, depending on whether the bank intends to hold them until maturity, these non-loan banking book assets can be either of:

1. **Fair Value Through Other Comprehensive Income (FVOCI)**,
2. **At Amortized Cost** or Held to Maturity (HTM).¹

For example, Treasury bonds, corporate bonds, MBS, etc., are usually FVOCI or HTM banking book assets.

FVOCI security

FVOCI assets in the banking book are typically debt securities that:

- Earn interest income for the bank.

¹Now called "at amortized cost" under IFRS 9 / AASB 9.

	Note	Group		Bank	
		30 Jun 23	30 Jun 22	30 Jun 23	30 Jun 22
		\$M	\$M	\$M	\$M
Assets					
Cash and liquid assets	5.1	116,619	161,154	108,367	150,974
Receivables from financial institutions	5.2	6,079	6,845	5,422	6,071
Assets at fair value through Income Statement	5.3	67,627	25,315	67,641	25,249
Derivative assets	5.4	23,945	35,736	25,585	37,774
Investment securities:					
At amortised cost	5.5	2,032	3,217	2,032	3,217
At fair value through Other Comprehensive Income	5.5	84,072	79,086	77,232	72,191
Assets held for sale	11.3	5	1,322	5	28
Loans and other receivables	3.1	926,082	878,854	816,140	773,042
Shares in and loans to controlled entities	11.2	–	–	54,636	56,719
Property, plant and equipment	6.1	4,950	4,887	3,549	3,627
Investments in associates and joint ventures	11.1	2,848	2,801	1,430	1,407
Intangible assets	6.2	7,393	6,899	4,340	3,883
Deferred tax assets	2.5	3,811	3,173	3,640	3,069
Other assets	6.3	7,382	5,971	6,799	5,387
Total assets		1,252,845	1,215,260	1,176,818	1,142,638

Figure 1: Excerpt of CBA's 2023 balance sheet - assets

- Can be sold before maturity but are *not actively traded* like FVTPL securities.
- Experience fair value changes that are recorded in **Accumulated Other Comprehensive Income (AOCI)** instead of directly impacting net income.
- Upon sale, unrealized gains/losses from AOCI are transferred to Net Income.

FVOCI security purchase

Assume that the bank now purchased a Treasury bond at a fair value of \$30,000. The bank may sell the bond before it matures, and hence classify it as a FVOCI asset on the banking book.

```
report5.print_balance_sheet()
```

Balance Sheet

Assets	
Cash and Cash Equivalents	\$31,359.00
Loans and Advances	\$80,000.00
Total assets	\$111,359.00
Liabilities	

Deposits and Other Public Borrowings	\$100,000.00
Deposits	\$100,000.00
Total liabilities	\$100,000.00
Net assets	\$11,359.00
Shareholders' equity	
Shareholders' Equity	\$10,000.00
Retained Earnings	\$1,359.00
Total shareholders' equity	\$11,359.00

Date: 2026-09-09

```

fvoci_security = MockInstrument("A Treasury bond (FVOCI)", BookType.BANKING_BOOK)
fvoci_security.value = 30_000
tx_fvoci_purchase = TransactionFactory.create_transaction(
    bank=bank,
    transaction_type=TransactionType.SECURITY_PURCHASE_FVOCI,
    instrument=fvoci_security,
    transaction_date=today,
    description="Purchase a FVOCI security",
)
bank.process_transaction(tx_fvoci_purchase)

report6 = Report(ledger=bank.ledger, viewer=viewer, date=today)
report6.print_balance_sheet()

```

Balance Sheet

Assets	
Cash and Cash Equivalents	\$1,359.00
Loans and Advances	\$80,000.00
Investment Securities	\$30,000.00
Investment Securities at FVOCI	\$30,000.00
Total assets	\$111,359.00
Liabilities	
Deposits and Other Public Borrowings	\$100,000.00
Deposits	\$100,000.00
Total liabilities	\$100,000.00

Net assets	\$11,359.00
Shareholders' equity	
Shareholders' Equity	\$10,000.00
Retained Earnings	\$1,359.00
Total shareholders' equity	\$11,359.00

Date: 2026-09-09

Note

The fair value of the security is recorded on the balance sheet under the *Investment Securities at FVOCI* account.

FVOCI security marking to market

- The *fair value* of FVOCI security is **marked to market** periodically, e.g., daily.
- This is because such securities do have market values (e.g., Treasury securities).

There are two possible cases:

1. Unrealized loss
2. Unrealized gain

Note

These gains and losses are *unrealized* because the securities have not yet been sold.

Important

These gains and losses are not through *trading*.

Unrealized gain

Suppose the next day the market value of the FVOCI security went up by 2% (\$600 = \$30,000 × 2%).

It is important to note that FVOCI security's fair value changes affect AOCI, an equity account.

It does not affect Income Statement P&L until it's eventually sold.

```
report6.print_balance_sheet()
```

Balance Sheet

Assets

Cash and Cash Equivalents	\$1,359.00
Loans and Advances	\$80,000.00
Investment Securities	\$30,000.00
Investment Securities at FVOCI	\$30,000.00
Total assets	\$111,359.00

Liabilities

Deposits and Other Public Borrowings	\$100,000.00
Deposits	\$100,000.00
Total liabilities	\$100,000.00

Net assets	\$11,359.00
------------	-------------

Shareholders' equity

Shareholders' Equity	\$10,000.00
Retained Earnings	\$1,359.00
Total shareholders' equity	\$11,359.00

Date: 2026-09-09

```
today = today + datetime.timedelta(days=1)
# Assume that the market value of the FVOCI security went up by 2%
new_fvoci_instrument_value = fvoci_security.value * 1.02
valuation_visitor = MockValuationVisitor(new_value=new_fvoci_instrument_value)
tx_fvoci_marked_up = TransactionFactory.create_transaction(
    bank=bank,
    transaction_type=TransactionType.SECURITY_FVOCI_MARK_TO_MARKET,
    instrument=fvoci_security,
    transaction_date=today,
    valuation_visitor=valuation_visitor,
    description="Mark to market a FVOCI security (up)",
)
bank.process_transaction(tx_fvoci_marked_up)

report7 = Report(ledger=bank.ledger, viewer=viewer, date=today)
report7.print_balance_sheet()
```

Balance Sheet

Assets	
Cash and Cash Equivalents	\$1,359.00
Loans and Advances	\$80,000.00
Investment Securities	\$30,600.00
Investment Securities at FVOCI	\$30,600.00
Total assets	\$111,959.00
Liabilities	
Deposits and Other Public Borrowings	\$100,000.00
Deposits	\$100,000.00
Total liabilities	\$100,000.00
Net assets	\$11,959.00
Shareholders' equity	
Shareholders' Equity	\$10,000.00
Accumulated OCI	\$600.00
Unrealized OCI Gain	\$600.00
Retained Earnings	\$1,359.00
Total shareholders' equity	\$11,959.00

Date: 2026-09-10

Unrealized loss

If the market value of the security went down by 2% ($\$612 = \$30,600 \times 2\%$) the next day, we recognize this *unrealized OCI loss* of AOCI.

```
report7.print_balance_sheet()
```

Balance Sheet

Assets	
Cash and Cash Equivalents	\$1,359.00
Loans and Advances	\$80,000.00
Investment Securities	\$30,600.00
Investment Securities at FVOCI	\$30,600.00
Total assets	\$111,959.00

Liabilities	
Deposits and Other Public Borrowings	\$100,000.00
Deposits	\$100,000.00
Total liabilities	\$100,000.00
Net assets	\$11,959.00
Shareholders' equity	
Shareholders' Equity	\$10,000.00
Accumulated OCI	\$600.00
Unrealized OCI Gain	\$600.00
Retained Earnings	\$1,359.00
Total shareholders' equity	\$11,959.00

Date: 2026-09-10

```

today = today + datetime.timedelta(days=1)
# Assume that the market value of the FVOCI security went down by 2%
new_fvoci_instrument_value = fvoci_security.value * 0.98
valuation_visitor = MockValuationVisitor(new_value=new_fvoci_instrument_value)
tx_fvoci_marked_down = TransactionFactory.create_transaction(
    bank=bank,
    transaction_type=TransactionType.SECURITY_FVOCI_MARK_TO_MARKET,
    instrument=fvoci_security,
    transaction_date=today,
    valuation_visitor=valuation_visitor,
    description="Mark to market a FVOCI security (down)",
)
bank.process_transaction(tx_fvoci_marked_down)

report8 = Report(ledger=bank.ledger, viewer=viewer, date=today)
report8.print_balance_sheet()

```

Balance Sheet

Assets	
Cash and Cash Equivalents	\$1,359.00
Loans and Advances	\$80,000.00
Investment Securities	\$29,988.00
Investment Securities at FVOCI	\$29,988.00
Total assets	\$111,347.00

Liabilities	
Deposits and Other Public Borrowings	\$100,000.00
Deposits	\$100,000.00
Total liabilities	\$100,000.00
Net assets	\$11,347.00
Shareholders' equity	
Shareholders' Equity	\$10,000.00
Accumulated OCI	(\$12.00)
Unrealized OCI Gain	\$600.00
Unrealized OCI Loss	\$612.00
Retained Earnings	\$1,359.00
Total shareholders' equity	\$11,347.00

Date: 2026-09-11

FVOCI, AOCI, and capital ratios

Because FVOCI's (even unrealized) gain/loss changes AOCI, and AOCI is part of a bank's CET1 capital, FVOCI's fair value movement can cause volatility in the bank's regulatory capital ratios.

Note

CET1 is the primary or core capital of a DI.

- **Common shares**
- Share premium resulting from the issue of CET1 instruments
- **Retained earnings**
- **Accumulated and other comprehensive income (AOCI)**
- Other disclosed reserves
- Certain minority interests
- Regulatory adjustment applied in the calculation of CET1 Capital.

In the U.S., banks with total assets below \$250 billion are permitted to exclude AOCI when reporting applicable regulatory ratios — the very opt-out that let SVB's unrealised bond losses stay invisible to its capital ratios.

FVOCI security sale

The next day, if the bank sold the FVOCI security at the prevailing market price (assumed 2% down), we recognize *realized OCI loss*.

Tracking the AOCI gain/loss of this particular FVOCI security,

1. On day 1, we had a gain of \$600.
2. On day 2, we had a loss of \$612.
3. On day 3, we had a loss of \$599.76.

The total net loss of this security is therefore \$611.76.

FVOCI security sale - Income Statement

Now that the FVOCI security is sold, its gain/loss is moved from AOCI to Income Statement as *Investment Income (FVOCI)*.

```
report8.print_income_statement()
```

Income Statement

Income	\$1,459.00
Interest Income	\$800.00
Trading Income (FVTPL)	\$659.00
Realized Trading Gain	\$1,159.00
Realized Trading Loss	\$500.00
Expense	\$100.00
Interest Expense	\$100.00
Profit	\$1,359.00

Date: 2026-09-11

```
from brms.models.transaction import SecuritySaleFVOCITransaction

today = today + datetime.timedelta(days=1)
# Assume that the market value of the FVOCI security went down by 2%
new_fvoci_instrument_value = fvoci_security.value * 0.98
valuation_visitor = MockValuationVisitor(new_value=new_fvoci_instrument_value)
tx_fvoci_marked_down = TransactionFactory.create_transaction(
```

```

    bank=bank,
    transaction_type=TransactionType.SECURITY_FVOCI_MARK_TO_MARKET,
    instrument=fvoci_security,
    transaction_date=today,
    valuation_visitor=valuation_visitor,
    description="Mark to market a FVOCI security (down)",
)
bank.process_transaction(tx_fvoci_marked_down)
# Sale transaction should be created after the previous transaction has been processed
tx_fvoci_sale = TransactionFactory.create_transaction(
    bank=bank,
    transaction_type=TransactionType.SECURITY_SALE_FVOCI,
    instrument=fvoci_security,
    transaction_date=today,
    description="Sale of FVOCI security (net loss)",
)
bank.process_transaction(tx_fvoci_sale)

report9 = Report(ledger=bank.ledger, viewer=viewer, date=today)
report9.print_income_statement()

```

Income Statement

Income	\$847.24
Interest Income	\$800.00
Trading Income (FVTPL)	\$659.00
Realized Trading Gain	\$1,159.00
Realized Trading Loss	\$500.00
Investment Income (FVOCI)	(\$611.76)
Realized OCI Gain	\$600.00
Realized OCI Loss	\$1,211.76
 Expense	 \$100.00
Interest Expense	\$100.00
 Profit	 \$747.24

Date: 2026-09-12

FVOCI security sale - Balance Sheet

When P&L accounts of income statement are closed, the net gain/loss of Investment Income (FVOCI) changes the bank's retained earnings.

```
report8.print_balance_sheet()
```

Balance Sheet

Assets

Cash and Cash Equivalents	\$1,359.00
Loans and Advances	\$80,000.00
Investment Securities	\$29,988.00
Investment Securities at FVOCI	\$29,988.00
Total assets	\$111,347.00

Liabilities

Deposits and Other Public Borrowings	\$100,000.00
Deposits	\$100,000.00
Total liabilities	\$100,000.00

Net assets	\$11,347.00
------------	-------------

Shareholders' equity

Shareholders' Equity	\$10,000.00
Accumulated OCI	(\$12.00)
Unrealized OCI Gain	\$600.00
Unrealized OCI Loss	\$612.00
Retained Earnings	\$1,359.00
Total shareholders' equity	\$11,347.00

Date: 2026-09-11

```
report9.print_balance_sheet()
```

Balance Sheet

Assets

Cash and Cash Equivalents	\$30,747.24
Loans and Advances	\$80,000.00
Total assets	\$110,747.24

Liabilities	
Deposits and Other Public Borrowings	\$100,000.00
Deposits	\$100,000.00
Total liabilities	\$100,000.00
Net assets	\$10,747.24
Shareholders' equity	
Shareholders' Equity	\$10,000.00
Retained Earnings	\$747.24
Total shareholders' equity	\$10,747.24

Date: 2026-09-12

Investment securities at amortized cost

- Measurement
 - Initially recorded at fair value plus transaction costs.
 - Subsequently measured at **amortised cost** using the **effective interest method**.
 - Carrying amount = purchase price $\pm$ cumulative amortisation of premium/discount
 - impairment allowance.
- Income Recognition
 - **Interest income** recognised in profit or loss using the effective interest rate (EIR).
 - Amortisation of premiums or discounts included in interest income.
- Gains and Losses
 - No recognition of unrealised fair value changes in P&L or OCI.
 - Realised gains/losses recognised in P&L only upon sale or derecognition.
- Impairment
 - Subject to **expected credit loss (ECL)** model under IFRS 9/AASB 9.
 - ECL recognised in P&L and reduces the carrying amount.